

I did not like the tiny font used by YouTube so I burned my subtitles into the video. Such subtitles become part of the video and cannot be turned off.

```
ffmpeg -y -i book-read.mp4 \
  -filter_complex
  "subtitles=transcript.ass" \
  -c:a copy \
  read-subtitled.mp4
```

While burning subtitles into the video is a failsafe alternative, your first option should be to add a subtitle stream (analogous to a video stream or an audio stream) in the multimedia file container. (MP4, MOV, MKV and other formats are just file containers. The actual codecs can be anything in the wild.) For Matroska videos, you can even add a custom subtitle display font file as a stream.

```
ffmpeg -i sarah.mp4 -i subtitle.ass \
  -map 0:v -map 0:a -map 1:s \
  -c:v copy -c:a copy -c:s ass \
  -metadata:s:s:0 language=eng \
  -attach Florentia.ttf \
  -metadata:s:3
  mimetype=application/x-truetype-font \
  sarah-subtitled.mkv
```

In this command, the `-map 1:s` ensures that the subtitles will be from the second file (the SubStation Alpha file). Even if the MP4 file had a



Figure 8: The SubStation Alpha (SSA) subtitle format offers the most features

subtitle stream, it will not be included in the MKV file.

SRT is the most popular subtitle format. SubStation Alpha (SSA) is the most feature-rich subtitle format because it supports a custom font, text colour, outline colour, shadow colour, background box and even rotation. All of this SSA awesomeness is supported only by the Matroska format.

The specification of the endearing but difficult SSA format is available on the <https://www.matroska.org> website (*Technical Info » Subtitles » SSA*). However, I will risk a description here for the style statement.

Style: Name, Fontname, Fontsize, PrimaryColour, SecondaryColour, OutlineColour, BackColour, Bold, Italic, Underline, StrikeOut, ScaleX, ScaleY, Spacing, Angle, BorderStyle, Outline, Shadow, Alignment, MarginL, MarginR, MarginV, Encoding

Name refers to a subtitle display style. You can define and use many different styles, not just the default. The colours are in hexadecimal AABBGRRR format! *PrimaryColour*

are automatically used when timestamps collide. *Bold*, *Italic*, et al, are -1 for true and 0 for false. (Yeah, I know.) *ScaleX* and *ScaleY* specify magnification (1-100). *Spacing* is additional pixel space between letters. *Angle* is about rotation (0-360) and controlled by *Alignment*. *BorderStyle* uses 1 (outline and drop-shadow) and 3 (outline box). If *BorderStyle* is 1, then *Outline* represents pixel space width (0-4) of its outline. In the same case, *Shadow* represents pixel space (0-4) below the text and shadow. *Alignment* takes 1 (left), 2 (centre) and 3 (right). If you add 4 to them, the subtitle appears at the top of the screen. If you add 8, it goes to the middle. Then, we have *Margin* from the left, right and bottom edges of the screen. *Encoding* is 0 for ANSI Latin and 1 for Unicode (I think).

I have a Caja Actions Configuration script that changes the style statement in a SSA file to:

```
Style: Default,Florentia,30,&H2
200CCCC,&H000000FF,&H2200 00EE,
&HAA00CCCC,-1,-1,0,0,100,100,0,
30.00,3,2,3,1,20,20,40,1. END
```



Figure 9: Default SSA styles used by FFmpeg, GNOME Subtitles and the author

is the colour of the subtitle text. *OutlineColour* is for the outline of the text. *BackColour* is the colour of the shadow behind the text. *SecondaryColour* and *OutlineColour*

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